TECHNICAL DESIGN: MULTI-SOURCE CANDIDATE TRANSFORMATION PIPELINE

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1. PIPELINE ARCHITECTURE

The candidate ingestion pipeline runs sequentially across decoupled phases to guarantee correctness, modularity, and determinism. Each stage executes separate business logic that protects the integrity of the data stream:

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Pipeline Flow: [Inputs] → (CSV / PDF Parsers) → [Raw Profiles] → (Normalizers) → [Normalized Profiles]
                        → (Merge Engine + Confidence Engine) → [Canonical Profile] → (Projector Layer) →
[Projected Candidate JSON] → (Validator) → [Valid Output]
```

- **Source Parsing:** Extracts structured data from CSV (Pandas) and unstructured text from PDF (pdfplumber) using heuristic-based layout parsing.
- **Normalization:** Sanitizes values (emails, phones, dates, skills, location components) to canonical standards prior to merging.
- **Merge & Confidence Engine:** Resolves field-level source conflicts and scores data confidence based on overlap and source trust.
- **Projection & Validation:** Filters, renames, and formats the output schema via a runtime JSON config, executing jsonschema validation before saving.

2. CANONICAL SCHEMA & NORMALIZATION RULES

- **candidate_id / full_name / headline:** Cleaned, whitespace-trimmed strings; name defaults to the first line of the resume.
- **emails / phones:** Emails are lowercased and deduplicated. Phones are normalized to the **E164** international format using the *phonenumbers* library.
- **location:** Parsed into city, state, and country. Country is standardized to **ISO-3166-1 alpha-2** format (e.g., 'IN' or 'US').
- **links:** Standardized list of URLs (LinkedIn, GitHub, LeetCode) prefixed with *https://*.
- **skills:** Match names against a canonical software engineering taxonomy using *rapidfuzz* similarity (threshold ≥ 80), keeping custom skills.
- **experience & education:** Nested lists. Dates normalized to **YYYY-MM** or 'Present'. Experience sorted newest-to-oldest by end date.
- **provenance:** Map of field paths to their source files (e.g., *recruiter.csv*, *resume.pdf*) and extraction methods (e.g., *pdf_extraction*, *csv_parsing*).

3. MERGE & CONFLICT RESOLUTION POLICY

If values overlap or conflict between sources, the engine applies deterministic priority and confidence adjustment rules:

- **Source Preference:** Recruiter CSV is preferred for contact details (emails, phones, location). Resume PDF is preferred for skill names, headline, experience, and education.
- **Confidence Base:** Resume PDF default confidence = **0.9**; Recruiter CSV default confidence = **0.8**.
- **Agreement Rule:** If sources agree on a field, confidence is boosted: C_merged = min(1.0, max(C_resume, C_csv) + 0.05) → e.g. 0.95 for name.
- **Conflict Rule:** If sources disagree, the preferred source's value is taken, but confidence is reduced: C_merged = C_preferred * 0.75 → e.g., 0.9 * 0.75 = 0.675.
- **Overall Confidence:** Computed as the simple average of all populated field confidences, providing a single quality metric for the profile.

4. RUNTIME PROJECTION & DYNAMIC VALIDATION

The **Projector Layer** reads a runtime configuration file to reshape the internal canonical record into the final candidate JSON structure. It supports subset selection, remapping paths (e.g., mapping index *emails[0]* to a flat key *primary_email*), executing field-level normalizations, toggling confidence/provenance blocks, and executing missing-value behaviors (*null*, *omit*, or raising a hard *error*). The **Validator Layer** translates the config's fields and types into a strict JSON Schema at runtime. It runs jsonschema validation and throws human-readable errors with precise violation paths if fields mismatch.

5. EDGE CASES & SCOPE DESCISION

- **Malformed CSV:** The parser handles empty files, missing columns, or parsing failures gracefully by logging errors and proceeding with remaining sources.
- **Missing Resume:** The pipeline runs end-to-end utilizing recruiter data only, mapping confidence to 0.8 and adjusting overall metrics.
- **Malformed Dates/Phones:** If standard parsing fails, fallback values (cleaned digits or raw text) are retained to prevent data loss, preserving original strings.
- **Out of Scope (Time/Resource Constraints):** OCR for scanned PDF documents (requires a readable text layer); deep NLP semantic analysis (replaced with robust regex and rapidfuzz matching); live external API calls for location lookup (replaced with mapping lists).